Segmented Phase Code Waveforms: A Novel Radar Waveform for Spaceborne MIMO-SAR

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***Abstract*— The echo separation issue associated with different transmit antennas is the most technical challenge in realizing the multiple-input multiple-output (MIMO) synthetic aperture radar (SAR) system. In this article, a novel MIMO-SAR imaging scheme based on an advanced radar waveform, namely, segmented-phase-code (SPC) waveform, is proposed. Compared with the state-of-the-art short-term shift-orthogonal (STSO) waveform beamforming schemes, this scheme relieves the short-term shift-orthogonality condition of transmitted waveforms without losing the imaging performance, which extends the optional waveform space. In this scheme, the separation of the echoes from close arrival angles is ensured by a simple time-shift weighting processing. Furthermore, a range bandpass filter bank and the digital beamforming (DBF) technique are employed to ensure that the echoes from far arrival angles are separable. Finally, detailed simulation experiments are performed to verify the feasibility of the proposed scheme, and in-depth discussions of different waveforms and MIMO-SAR imaging schemes are presented.**

***Index Terms*— Digital beam forming (DBF), echo separation, multiple-input multiple-output (MIMO) synthetic aperture radar (SAR) (MIMO-SAR), segmented-phase-code (SPC) waveform.**

I. Introduction

SYNTHETIC aperture radar (SAR) is a powerful remote-sensing technique for global comprehensive environmental monitoring, day-and-night and all-weather surveillance. Expanding application requirements and increasingly stringent technical requirements, such as wider swath width and higher resolution, have driven research on new SAR system concepts and the multiple-input multiple-output (MIMO) SAR concept is emerged.

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The MIMO concept was first proposed in the wireless communication field to increase the channel capacity [1] and it has displayed excellent anti-noise and anti-narrowband fading ability [2], [3]. The multiple antenna concept in the radar field, called MIMO radar, has been triggering active research work and is gaining popularity because of the vast potential for improving radar performance [4]. As a type of active imaging radar system, MIMO-SAR is also very attractive because it can provide more degrees of freedom than traditional systems and dramatically improve SAR system performance [5]–[7]. For example, MIMO-SAR can produce more spatial samples in the azimuth direction and longer baselines, making MIMO-SAR attractive for applications such as high-resolution wide-swath (HRWS) SAR imaging [8], ground moving target indication (GMTI) [9], [10], and interference and tomography applications [11]. In addition, MIMO-SAR has inspired a new downward-looking 3-D radar imaging concept called ARTINO, of which the cross-track resolution was improved [12], [13]. Furthermore, compared with the multistatic SAR, MIMO-SAR can obtain more images of the same area acquired from slightly different incident angles, so it can achieve a higher resolution in range by jointly exploiting both mono- and bistatic acquisitions [14], [15].

In the MIMO-SAR system, each antenna receives the ground scattered echoes from all the transmit antennas, and for the echo signal of a transmit antenna, the echoes from other transmit antennas are interference signals that can considerably degrade the quality of radar image. Therefore, the separation for the echoes from different transmit antennas is the most challenging issue related to employing the MIMO-SAR concept.

A. Previous Work

Generally, MIMO-SAR echo separation schemes can be divided into the following components: 1) time-division multiplexing (TDM) waveforms [16], [17]; 2) frequency-division multiplexing (FDM) waveforms [18], [19]; and 3) code-division multiplexing (CDM) waveforms [20], [21]. The TDM waveform scheme will increase the total pulsewidth due to time division, which will greatly reduce the swath width, and the FDM waveform scheme will reduce the coherence and increase the system costs because it requires extra mixing circuits to obtain the radar signals in different frequency bands. Therefore, the applications of the TDM and FDM waveform schemes have been extremely limited in spaceborne

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MIMO-SAR, and the CDM waveform scheme is a more promising scheme.

In the CDM waveform scheme, the initial approach involves using mutually orthogonal waveforms to separate echoes in the time-frequency domain [21], [22]. However, Krieger [7] noted that these suggested orthogonal waveforms cannot meet the requirements of MIMO-SAR considering Parseval's theorem[1] and the distributed target features of SAR images.[2] To solve this dilemma, multidimensional coding waveforms have been suggested. The space-time coding (STC) Alamouti schemes, which code on the transmitter and decode on the receiver, was proposed by Kim [23] and Wang [21] *et al*. This approach was further researched for suiting the time-variant channel response [24], [25] and improving the signal-to-noise ratio (SNR) of the echoes [26].

In addition, orthogonal frequency-division multiplexing (OFDM) chirp waveforms represent a promising candidate for suppressing interference energy because of the orthogonality of discrete frequency components. However, the orthogonality condition can only be satisfied at a certain length. To solve this problem, Kim *et al.* [27] employed the digital beam forming (DBF) technique to divide a swath into multiple sub-swaths, so as to reduce the echo length to meet the orthogonality requirement, and the improved scheme was proposed in [28].

On the other hand, OFDM waveforms suffer from the effect of Doppler leakage, which will lead a deterioration of the separation performance. In this respect, the orthogonal-waveform beamforming scheme, which combines orthogonal waveforms and DBF technique, was suggested by Krieger [7]. In this scheme, the mutual interference from the closely spaced scatterers can be avoided using a class of short-term shift-orthogonal (STSO) waveforms while the interferences from more distant scatterers can be suppressed by DBF on receive. Furthermore, the segmental-shifted chirp (SSC) waveform beamforming scheme was introduced in [29], and it will be verified by a real MIMO-SAR system architecture using the airborne radar sensor DBSAR-2 operated by NASA.

B. Main Advantages and Organization

In this article, a novel spaceborne MIMO-SAR imaging scheme based on segmented-phase-code (SPC) waveforms, which suits for M transmitters, is proposed to avoid the echo interference of different antennas. In this scheme, the transmitted waveform is divided by some signal segments and well-designed phase values are used to code a modulation waveform to construct each signal segment. The modulation waveform is arbitrary and it decides the range response after echo separation. After receiving and recording ground reflected echoes, a simple time-shift weighting processing is employed to separate the echoes from close arrival angles. Finally, a range bandpass filter bank and DBF technique are employed to separate the echoes from far arrival angles.

[1]The mismatched energy of orthogonal waveforms with flat spectra is spread over the image but not removed, see the APPENDIX A of [7].

[2]The impact of the mismatched energy cannot be neglected because the mismatched energy from numerous targets will be accumulated.

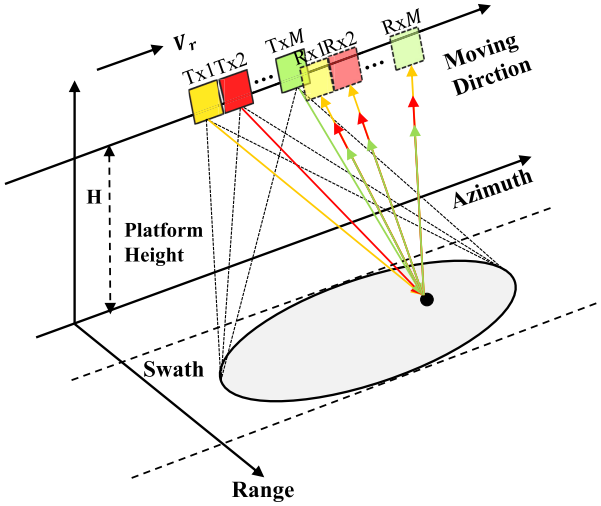


Fig. 1. Geometry illustration of an MIMO-SAR system.

Compared with the existing schemes, the main advantages of the proposed scheme are as follows.

1) Compared with the state-of-the-art STSO waveform beamforming scheme, the proposed scheme can relieve the short-term shift-orthogonality condition without losing the imaging performance, which extends the optional waveform space.
2) Compared with the other existing MIMO-SAR schemes, the proposed method avoids the effects of Doppler leakage and suits for time-varying channel responses, and it does not need to increase pulse-repetition frequency (PRF).

The rest of this article is organized as follows. In Section II, first, the echo separation problem is reviewed, then, the SPC waveform and spaceborne MIMO-SAR imaging scheme are introduced. In Section III, the effects of swath width, bandpass filter numbers are detailed, and the optimal phase coding scheme for a maximum SNR is given. Furthermore, the detailed simulation experiments are performed to verify the practicability of the proposed scheme in Section IV. In Section V, the performance comparisons between the proposed method and the existing MIMO-SAR imaging schemes are detailed. Finally, Section VI concludes this article with a brief discussion.

Notation: Boldface uppercase and lowercase letters denote matrices and vectors, respectively; $(\cdot)^T$, $(\cdot)^H$ denote transpose and Hermitian transpose, respectively; $*_t$ denotes the fast time convolution; Rank($\boldsymbol{A}$) is the rank of matrix $\boldsymbol{A}$; $\boldsymbol{A}_{n,m}$ refers the element in the nth row and mth column; finally, $\boldsymbol{I_k}$ indicates the identity matrix with size $k \times k$.

II. MIMO-SAR Echo Separation Scheme

A. Simple Overview of the Echo Separation Problem

Fig. 1 shows geometry illustration of an MIMO-SAR system, and it is worth noting that the differences between the position of Tx/Rx antennas are magnified. In the go-stop model, the positions of Tx/Rx antennas are the same, and to obtain as many equivalent phase centers as possible,

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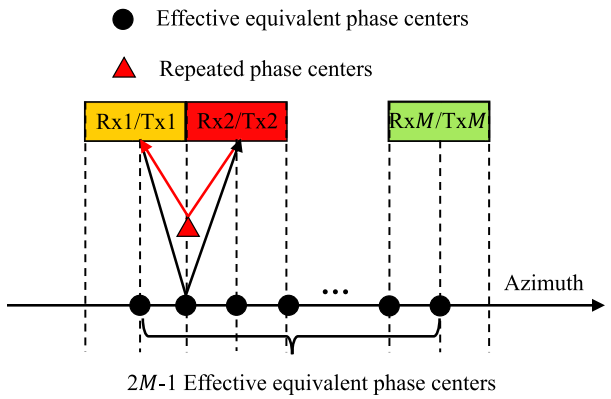


Fig. 2. Equivalent phase centers provided by an MIMO-SAR system with M Tx/Rx antennas.

MIMO-SAR system uses all Tx/Rx antennas to transmit radar signals and simultaneously receive the ground scattered echoes. For a multichannel SAR system, if M Tx/Rx antennas are employed, the maximum number of possible equivalent phase centers is M, and this number can increase to M^2 for an MIMO-SAR system. However, there are some repeated equivalent phase centers. For example, as shown in Fig. 2, the acquired phase center from Rx1 to Tx2 is the same as that from Tx1 to Rx2. Therefore, the effective equivalent phase center of an MIMO-SAR system with M Tx/Rx antennas is reduced to $2M-1$, which suggests that MIMO-SAR system can have a lower PRF, wider swath, additional, and longer baselines [7].

However, the scattered echoes from different transmit antennas are received simultaneously by each receive antenna and hereinafter, we assume that the radar signal s_m is transmitted by the mth antenna and it can be expressed as

$$s_m(t) = \text{rect}\left(\frac{t}{T}\right)\exp(j\theta_m(t)) \tag{1}$$

where t is the fast time, T is the pulselength, $\theta_m(t)$ is the phase of mth transmitted radar signal, and

$$\text{rect}(t) = \begin{cases} 1, & -0.5 \le t \le 0.5 \\ 0, & \text{elsewhere.} \end{cases} \tag{2}$$

After being reflected by a point target, the echo signal is down converted to the baseband, and the echo signal transmitted by the mth antenna and received by the pth antenna can be expressed as

$$r_{p,m}(t,\eta) = \sigma_0 \text{rect}\left(\frac{t-t_{p,m}}{T}\right)\exp(-j2\pi f_c t_{p,m}) \times \exp(j\theta_m(t-t_{p,m})) \tag{3}$$

where σ_0 is the backscattering coefficient, and f_c is the carrier frequency. In addition, $t_{p,m}$ is the round-trip delay of the signal, which can be further expressed as

$$t_{p,m} = \frac{R_{p,m}(\eta)}{c} \tag{4}$$

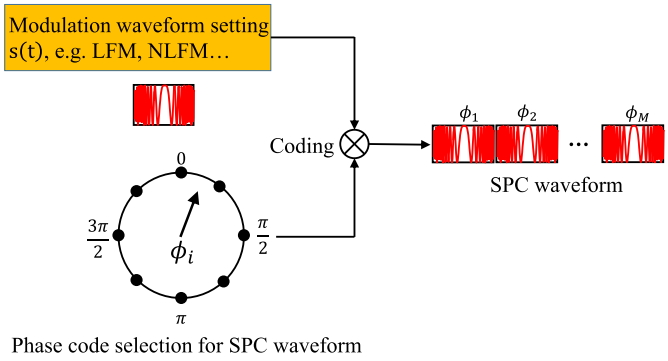


Fig. 3. Notional representation of the generation of a SPC waveform.

where c is the speed of light, and $R_{p,m}(\eta)$ represents the slant distance history which is a function of the azimuth time η, as

$$\begin{aligned} R_{p,m}(\eta) &= R_m(\eta) + R_p(\eta) \\ &= \sqrt{R_0^2 + \left(V_r\eta + \left(m - \frac{M+1}{2}\right)\frac{L_a}{M}\right)^2} \\ &\quad + \sqrt{R_0^2 + \left(V_r\eta + \left(p - \frac{M+1}{2}\right)\frac{L_a}{M}\right)^2} \end{aligned} \tag{5}$$

where $R_m(\eta)$ denotes the slant range history between the mth antenna and the target, R_0 is the closest slant range that uses the azimuth antenna center $((M+1)/2)$ as the reference position, η is the slow time referenced to the time of closest approach, V_r is the speed of platform, and L_a is the azimuth antenna length. Furthermore, (3) can be represented as a more simple form

$$r_{p,m}(t,\eta) = h_{p,m}(t,\eta) *_t s_m(t) \tag{6}$$

where $h_{p,m}(t,\eta)$ denotes the channel response corresponding to the mth transmitter and the pth receive, as

$$h_{p,m}(t,\eta) = \sigma_0\delta(t - t_{p,m}(\eta))\exp(-j2\pi f_c t_{p,m}(\eta)). \tag{7}$$

The mixed echo received by the pth antenna can be expressed as

$$r_p(t,\eta) = \sum_{m=1}^{M} r_{p,m}(t,\eta) \tag{8}$$

and the goal of the echo separation processing is to obtain the echo of each transmit antenna, namely, $r_{p,1}(t), r_{p,2}(t), \ldots, r_{p,M}(t)$.

B. Scheme Description

The main objective of the orthogonal-waveform beamforming separation scheme can be summarized as two processing steps. The first step is to separate the echoes from close arrival angles. Then, a spatial filter (namely, DBF technique) is employed to separate the echoes from far arrival angles. To achieve the separation goal of the first step, well-designed orthogonal waveforms, for example, OFDM waveforms, are employed. However, the Doppler shift of the radar will significantly degrade the waveform orthogonality and affect the separation performance, which is detailed in [7], [30], and [31].

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In this respect, STSO waveforms are proposed to avoid the effect of Doppler leakage, whereas the transmitted waveforms should meet the short-term shift-orthogonality condition. In the proposed scheme, the orthogonality of transmitted waveforms is not the key for echo separation, and a new advanced radar waveform named the SPC waveform is introduced. The simple construction of the SPC waveform is shown in Fig. 3 and it can be formulated as

$$s_{\mathrm{spc}}(t) = \sum_{i=1}^{M} \phi_i s(t-(i-1)T_s) \tag{9}$$

where $s(t)$ is a subpulse signal that used as a modulation waveform and can be an arbitrary radar signal, for example, a linear frequency modulation (LFM) waveform or a nonlinear frequency modulation (NLFM) waveform [32], [33]. In (9), ϕ_i is a constant phase coding ith signal segment. If the pulse duration and bandwidth of $s(t)$ are T_s and B, respectively, the corresponding pulselength and bandwidth of the SPC waveform are $T = MT_s$ and B.

Furthermore, considering an MIMO-SAR system with M transmit antennas, the implemented SPC waveform of mth antenna can be represented as

$$s_m(t) = \sum_{i=1}^{M} \phi_{m,i} s(t-(i-1)T_s) \tag{10}$$

where $\phi_{m,i}$ denotes the phase code of the ith segment of the mth transmit antenna. Considering the form of transmitted waveforms, (6) can be further expressed as

$$r_{p,m} = h_{p,m}(t,\eta) *_t \boldsymbol{\phi_m s} \tag{11}$$

where $\boldsymbol{\phi_m}$ denotes the phase code vector of the mth transmitted waveform, as

$$\boldsymbol{\phi_m} = [\phi_{m,1}, \phi_{m,2}, \ldots, \phi_{m,M}] \tag{12}$$

and $\boldsymbol{s}$ is the modulation waveform vector

$$\boldsymbol{s} = [s(t), s(t-T_s), \ldots, s(t-(M-1)T_s)]^T. \tag{13}$$

The mixed echo $r_p(t,\eta)$ can be further represented as

$$r_p(t,\eta) = \sum_{m=1}^{M} r_{p,m}(t) = \boldsymbol{h_p} *_t \boldsymbol{\Phi s} \tag{14}$$

where $\boldsymbol{h_p}$ is the channel response vector

$$\boldsymbol{h_p} = [h_{p,1}, h_{p,2}, \ldots, h_{p,M}] \tag{15}$$

and $\boldsymbol{\Phi}$ is the phase code matrix

$$\boldsymbol{\Phi} = \begin{bmatrix} \boldsymbol{\phi_1} \\ \boldsymbol{\phi_2} \\ \vdots \\ \boldsymbol{\phi_M} \end{bmatrix}. \tag{16}$$

To obtain the pth receive antenna echo of the transmitted signal $s_m(t)$, namely, $r_{p,m}(t)$, a simple time-shift weighting processing is used for the mixed echo signal $r_p(t)$, as

$$\begin{aligned} r_{p,m}^{\mathrm{weight}}(t,\eta) &= \sum_{i=1}^{M} w_{m,i} r_p(t-(i-1)T_s,\eta) \\ &= \sum_{i=1}^{M} w_{m,i} \boldsymbol{h_p} *_t \boldsymbol{\Phi s_i} \end{aligned} \tag{17}$$

where $r_p(t-(i-1)T_s,\eta)$ $(i=1,2,\ldots,M)$ denotes the ith time-shifted echo signal, and $\boldsymbol{s_i}$ is the time-shift modulation waveform vector, as

$$\boldsymbol{s_i} = \begin{bmatrix} s(t-(i-1)T_s) \\ s(t-(i-1)T_s) \\ \vdots \\ s(t-(M-1)T-(i-1)T_s) \end{bmatrix}. \tag{18}$$

Obviously, the time-shifted echo signal have a common time segment in $(M-1)T_s+t_{p,m} \le t \le MT_s+t_{p,m}$. In other word, time-shifted echo signals have the common term, namely, $s(t-(M-1)T_s)$. The time-shifted echo signal in the time period of $(M-1)T_s + t_{p,m} \le t \le MT_s + t_{p,m}$ can be further represented as

$$\begin{aligned} & r_{p,m}^{\mathrm{common}}(t,\eta) \\ &= \boldsymbol{h_p} *_t \boldsymbol{\Phi} \left\{ \begin{bmatrix} 0 \\ 0 \\ \vdots \\ w_{m,1} s(t-(M-1)T_s) \end{bmatrix} \right. \\ &\qquad + \begin{bmatrix} 0 \\ \vdots \\ w_{m,2} s(t-(M-1)T_s) \\ 0 \end{bmatrix} \\ &\qquad \left. + \cdots + \begin{bmatrix} w_{m,M} s(t-(M-1)T_s) \\ \vdots \\ 0 \\ 0 \end{bmatrix} \right\} \\ &= \boldsymbol{h_p} *_t \{ s(t-(M-1)T_s) \boldsymbol{\Phi w_m} \}, \end{aligned} \tag{19}$$

where $\boldsymbol{w_m} = [w_{m,M}, w_{m,M-1}, \ldots, w_{m,1}]^T$ denotes the echo signal weighting coefficient vector for the separation of echo signal $r_{p,m}(t,\eta)$. Moreover, $\boldsymbol{\Phi}$ and $\boldsymbol{w_m}$ are not the function of the variable t, thus, (19) can be recast as

$$r_{p,m}^{\mathrm{common}}(t,\eta) = s(t-(M-1)T_s) *_t \boldsymbol{h_p \Phi w_m}. \tag{20}$$

Obviously, $\boldsymbol{w_m}$ should satisfy the following condition:

$$\boldsymbol{w_m} = \boldsymbol{\Phi}^{-1} \boldsymbol{c_m} \tag{21}$$

where $\boldsymbol{c_m}$ denotes the expected weighted result in $(M-1)T_s + t_{p,m} \le t \le M \cdot T_s + t_{p,m}$ for separating echo $r_{p,m}(t,\eta)$, as

$$\boldsymbol{c_m} = [\underbrace{0,\ldots,0}_{m-1}, 1, \ldots, 0]^T. \tag{22}$$

As shown in (21), to obtain an effective weighting coefficient vector $\boldsymbol{w_m}$, the phase code matrix $\boldsymbol{\Phi}$ must be invertible. In other word, for the signals in $(M-1)T_s + t_{p,m} \le t \le MT_s + t_{p,m}$, utilizing the well-designed weighting coefficient

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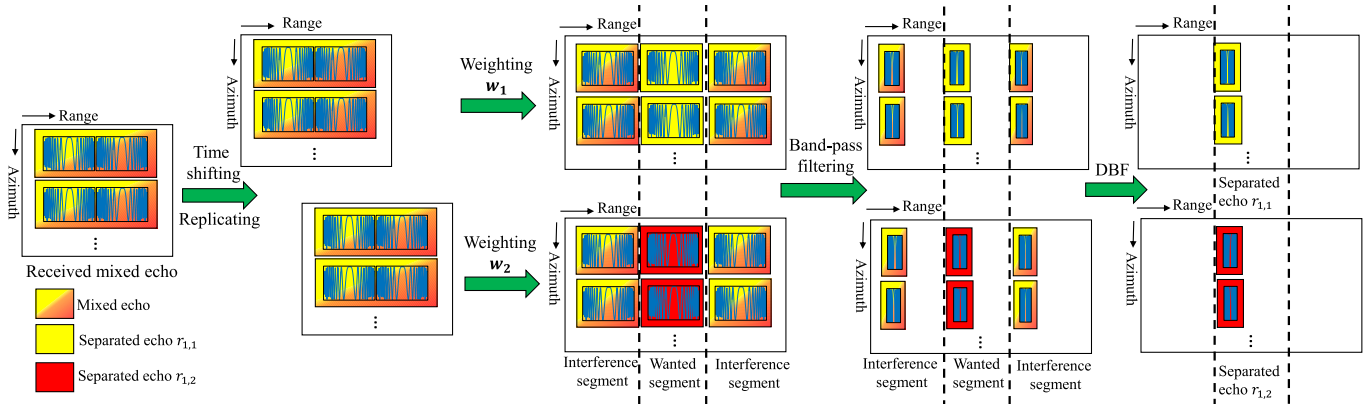


Fig. 4. Illustration of the echo separation processing for the 1st receive antenna ($M = 2$), where the range migration is ignored. Besides, only a single bandpass filtering result for a receive channel echo is given, and the full-bandwidth results can be stitched by all filtering results.

vector $\boldsymbol{w_m}$, $r_{p,m}(t)$ will be preserved while the interference signals will be removed, namely,

$$r_{p,m}^{\text{weight}}(t,\eta) = r_{p,m}(t,\eta) \quad (M-1)T_s + t_{p,m} \le t \le MT_s + t_{p,m}. \tag{23}$$

Considering $M = 2$ transmit antennas, a simple illustration of the echo separation processing is shown in Fig. 4. After time-shift weighting processing, the separated echo $r_{p,m}$ can be obtained in the common time segment. Then DBF is employed to suppress the interference echoes from far arrival angles. However, because of the time continuity of the wanted signal segment and the interference signal segment, a unique separation cannot be provided with DBF, and additional band-pass filters are needed to decrease pulse extension loss (PEL). After bandpass filtering, the wanted echo and the interference echoes are separated in the time domain as several narrowband signal segments, and the interference echoes can be suppressed by DBF. The role of bandpass filter is equivalent to that of the matched filter mentioned in [7]. However, the bandpass filter is much easier to employ than matched filter, and it can greatly reduce the cost of the spaceborne MIMO-SAR system.

Furthermore, Fig. 5 shows the generic MIMO-SAR solution scheme diagram based on SPC waveforms and the schematic of antenna unit is shown in Fig. 6. Assume that the number of the antennas in azimuth is M_{A} and the number of the antennas in elevation is M_{E}. As discussed before, the antennas in azimuth transmit different SPC waveforms and all antennas simultaneously receive the ground echo, resulting in $M_{\text{A}} \times M_{\text{E}}$ echo data streams. Then, the echo date are replicated, time-shifted and weighted, leading to $M_{\text{A}} \times M_{\text{E}} \times M_{\text{A}}$ echo data streams. Next, bandpass filters are employed, yielding several narrowband echo data segments. Finally, DBF is performed for all bandpass filtering results, and the narrowband echo data segments after DBF are reconstructed to full band echo data, resulting in $M_{\text{A}} \times M_{\text{A}}$ separated echo data streams. Besides, MIMO-SAR echo separation processing can be implemented in parallel for each receive channel. After completing the echo separation, the echo data can be fed to the conventional SAR image processing, which yields $M_{\text{A}} \times M_{\text{A}}$ SAR images. In addition, for different applications, some corresponding post-processing steps can be employed.

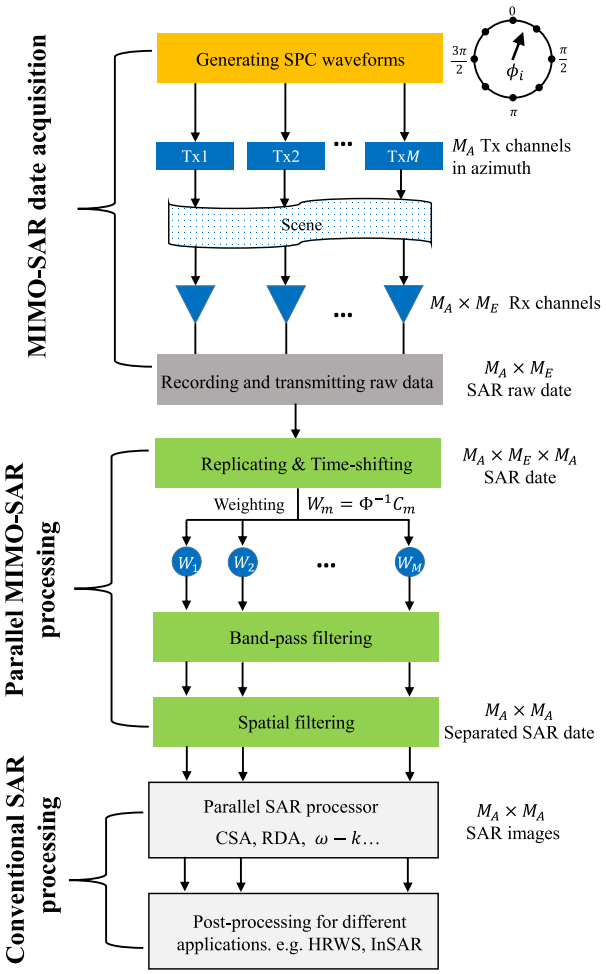


Fig. 5. Generic MIMO-SAR solution scheme based on SPC waveforms.

C. Spatial Filter Design

In this section, the design of the spatial filter is given in detail. After performing time-shift weighting, N_{B} bandpass filters are used to divide the echo into several time segments. Here, without loss of generality, up chirp waveform is used as the modulation waveform of SPC waveforms, and the bandpass filtering result for No. n_{B} filter can be represented

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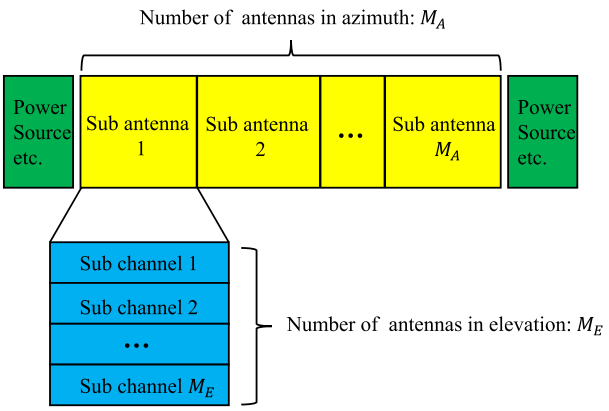


Fig. 6. Schematic of the antenna unit.

as

$$r_{\mathrm{BF}}(t,\eta) = \mathrm{rect}\left(\frac{t-t_{p,m}-T_{\mathrm{want}}(n_B)}{\frac{T_s}{N_B}}\right) r_{p,m}^{\mathrm{weight}}(t,\eta) + \sum_{i=-M+1,i\neq 0}^{M-1} \mathrm{rect}\left(\frac{t-t_{p,m}-T_{\mathrm{want}}(n_B)-iT_s}{\frac{T_s}{N_B}}\right) \times r_{p,m}^{\mathrm{weight}}(t,\eta) = r_{\mathrm{want}}(t,\eta) + r_{\mathrm{inf}}(t,\eta) \quad (24)$$

where $T_{\mathrm{want}}(n_B)$ denotes the time shift referenced to the echo pulse center of the wanted signal segment, and it can be represented as

$$T_{\mathrm{want}}(n_B) = \frac{(M-2)T_s}{2} + \frac{(2n_B-1)T_s}{2N_B}. \quad (25)$$

Besides

$$r_{\mathrm{want}}(t,\eta) = \mathrm{rect}\left(\frac{t-t_{p,m}-T_{\mathrm{want}}(n_B)}{\frac{T_s}{N_B}}\right) r_{p,m}^{\mathrm{weight}}(t,\eta) \quad (26)$$

and according to (23), $r_{p,m}^{\mathrm{weight}}(t) = r_{p,m}(t)$ in $(M-1)T_s + t_{p,m} \le t \le MT_s + t_{p,m}$. Hence, (26) can be reformulated as

$$r_{\mathrm{want}}(t,\eta) = \mathrm{rect}\left(\frac{t-t_{p,m}-T_{\mathrm{want}}(n_B)}{\frac{T_s}{N_B}}\right) r_{p,m}(t,\eta) \quad (27)$$

which represents the filtered out result of the wanted signal, and

$$r_{\mathrm{inf}}(t,\eta) = \sum_{i=-M+1,i\neq 0}^{M-1} \mathrm{rect}\left(\frac{t-t_{p,m}-T_{\mathrm{want}}(n_B)-i\times T_s}{\frac{T_s}{N_B}}\right) \times r_{p,m}^{\mathrm{weight}}(t,\eta) \quad (28)$$

is the interference echo signal that should be suppressed by the spatial filter. The actual direction of arrival of the echo is dependent on the slant range of the target, namely, the echo time delay. And the time delay difference between the wanted echo and the ith interference echo is iT_s. Therefore, the wanted and interference echoes are received from different elevation angles, respectively. Considering a DBF system with N sub-antennas in elevation, the steering vectors of the wanted echoes and the interference echoes can be represented as

$$\boldsymbol{v_0}(k) = [e^{jd_1k_0}, \ldots, e^{jd_Nk_0}]^T \quad (29)$$

respectively, and

$$\boldsymbol{v_i}(k) = [e^{jd_1k_i}, \ldots, e^{jd_Nk_i}]^T \quad i = [-M+1, \ldots, M-1] \quad i \neq 0 \quad (30)$$

where $\lambda = c/f_c$ is the wavelength, $d_n = (n-((N+1)/2))d$, d is the channel interval in elevation, and N denotes the channel number of DBF. Besides, k_0 is the wavenumber of the wanted echo

$$k_0 = \frac{2\pi}{\lambda}\sin(\varphi_0) \quad (31)$$

and k_i is the wavenumber of the ith interference echo

$$k_i = \frac{2\pi}{\lambda}\sin(\varphi_i) \quad (32)$$

where φ_0 and φ_i are the off-boresight angles corresponding to the time delay terms in (26) and (28), respectively. In this article, the DBF based on the linear constraint minimum variance (LCMV)[3] [34] is employed. And the matrix of steering vectors can be constructed as

$$V = \left[\boldsymbol{v_0}, \boldsymbol{v_{-M+1}}, \ldots, \boldsymbol{v_{M-1}}, \frac{d\boldsymbol{v_{-M+1}}}{dk_{-M+1}}, \ldots, \frac{d\boldsymbol{v_{M-1}}}{dk_{M-1}}\right] \quad (33)$$

where

$$\frac{d\boldsymbol{v_i}}{dk_i} = [jd_1e^{jd_1k_i}, \ldots, jd_Ne^{jd_Nk_i}]^T \quad i = [-M+1, \ldots, M-1], \quad i \neq 0 \quad (34)$$

The weight coefficient can be calculated as

$$\boldsymbol{w}_{\mathrm{DBF}} = \boldsymbol{g}(\boldsymbol{V}^H\boldsymbol{V})^{-1}\boldsymbol{V}^H \quad (35)$$

where $\boldsymbol{g} = [1, 0, \ldots, 0]$ (the number of zeros is $4M-4$) is the constraint vector.

III. Performance Analysis

In this section, the effects of the swath width and the number of the bandpass are discussed. Moreover, the optimal phase coding scheme for a maximum SNR is given.

A. Swath Width

First, consider a scene with a swath width shorter than the spatial extension of the sub-pulsewidth [namely, $(T_sc/2)$], the scene echo after range match filtering and time-shift weighting can be represented as

$$r_{\mathrm{scene}}(t) = \int_{t_{\min}}^{t_{\min}+W_s} \sum_{i=1}^{M} w_{m,i}\, Q_p(t-t_{p,m}-(i-1)T_s)dt_{p,m} \quad (36)$$

where $Q_p(t)$ is the impulse response of $r_p(t)$. It should be noted that $Q_p(t)$ is consisted of a series of point target impulse

[3]For a better suppression effect, the first-order derivative constraints for the interference steering vectors are considered.

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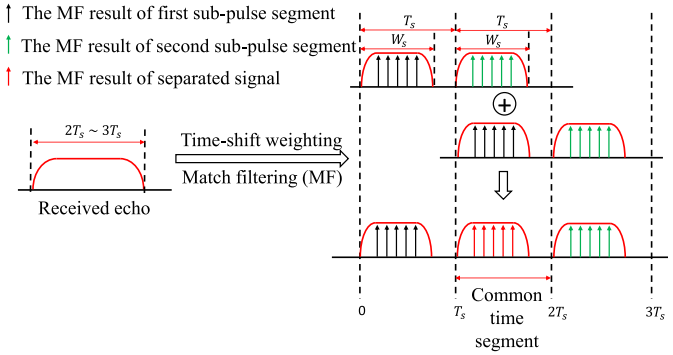


Fig. 7. Illustration of echo separation processing in the case of $M = 2$ and the swath width shorter than $(T_s c/2)$.

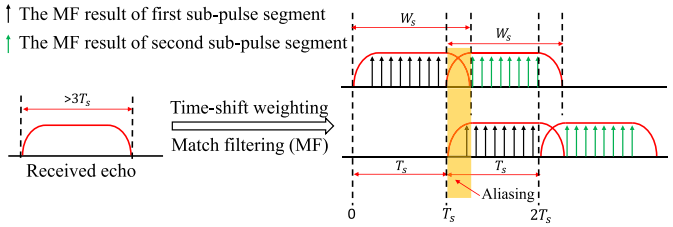


Fig. 8. Illustration of echo separation processing in the case of $M = 2$ and the swath width longer than $((T_s \cdot c)/2)$.

response functions, whereas the detailed mathematical expression of $Q_p(t)$ is not important for the analysis. In addition, $t_{\min}$ ($t_{\min} + W_s$) is the delay of nearest (farthest) range point. Moreover, due to the swath width assumption of $W_s < T_s$, (36) can be reformulated as

$$r_{\text{scene}}(t) = \int_{t_{\min}}^{t_{\min}+W_s} Q_{p,m}(t - t_{p,m} - (i-1)T_s)dt_{p,m} \quad (M-1)T_s + t_{\min} \le t \le MT_s + t_{\min} \tag{37}$$

where $Q_{p,m}(t)$ is the impulse response of the $r_{p,m}(t)$. That is to say, the mixed echoes for a short swath scene can be successfully separated without additional DBF. A simple explanation of $M = 2$ in this situation is shown in Fig. 7. It can be seen that only one type of signal segment exists in the common time segment. Then, the echoes in the common time segment are weighted for separation. Similarly, DBF is also unnecessary for other already proposed MIMO-SAR solutions [7], [27], [30], [31] because the orthogonality of the OFDM waveforms can be guaranteed when swath width is shorter than $(T_s c/2)$.

However, if the swath width is longer than $(T_s c/2)$, more than one type signal segment will exist in the common time segment, and the aliasing signal will destroy the demodulation condition, which will lead to the unsuccessful separation. A simple explanation of $M = 2$ in this situation is shown in Fig. 8. As shown in the yellow rectangle in Fig. 8, the tail of the echo of the first sub-pulse is aliased with the beginning of the second sub-pulse echo. This aliasing will lead to the failure of the echo separation. Generally, the swath width is larger than the sub-pulsewidth for spaceborne SAR, therefore, DBF is necessary in spaceborne MIMO-SAR systems.

B. SNR Performance

In this section, the effect of noise is taken into account, and the design of the optimal phase code matrix is discussed.

The received mixed echo ($\tilde{r}_p(t, \eta)$) with an additive noise $n_p(t, \eta)$ can be represented as

$$\tilde{r}_p(t, \eta) = r_p(t, \eta) + n_p(t, \eta). \tag{38}$$

Following the discussion in Section II-A [see (23)] and considering the echo separation processing for $\tilde{r}_p(t, \eta)$, the time-shifted and weighted echo signal in the common time segment is given by

$$\hat{r}_{p,m}^{\text{weight}}(t, \eta) = r_{p,m}(t, \eta) + \boldsymbol{n'_p}(t, \eta)\boldsymbol{w_m} \tag{39}$$

where

$$\boldsymbol{n'_p}(t, \eta) = \left[n'_{p,1}(t, \eta), n'_{p,2}(t, \eta), \ldots, n'_{p,M}(t, \eta)\right] \tag{40}$$

and $n'_{p,i}(t, \eta)$ ($i = \{1, 2, \ldots, M\}$) denotes the ith time-shifted noise in the common time segment

$$n'_{p,i}(t, \eta) = n_p(t - (i-1)T_s, \eta) \quad (M-1)T_s + t_{p,m} \le t \le M \cdot T_s + t_{p,m}. \tag{41}$$

It is worth noting that $n'_{p,i}(t, \eta)$ is independent identically distributed because they are the different time segment of $n_p(t, \eta)$. To increase SNR, the design of optimal weighting coefficient vector can be formulated as the following constrained optimization:

$$\begin{cases} \min\limits_{\boldsymbol{w_m}} \sum\limits_{i=1}^{M} |w_{m,i}|^2 \\ \text{s.t. } \boldsymbol{\phi_m w_m} = 1, \quad \boldsymbol{w_m} \in \mathbb{C}^M \end{cases} \tag{42}$$

where the constraint condition $\boldsymbol{\phi_m w_m} = 1$ can be obtained from (21). Considering the following inequation (namely, Cauchy-Schwarz inequation):

$$1 = \left|\sum_{i=1}^{M} \phi_{m,i} w_{m,i}\right|^2 \le \sum_{i=1}^{M} |\phi_{m,i}|^2 \sum_{i=1}^{M} |w_{m,i}|^2 = M \sum_{i=1}^{M} |w_{m,i}|^2. \tag{43}$$

That is to say, when the optimal weighting coefficient vector are set as

$$\boldsymbol{w_m^{\text{opt}}} = \frac{1}{M}\boldsymbol{\phi_m}^H \tag{44}$$

the minimum value of the goal can be obtained, namely,

$$\sum_{i=1}^{M} \left|w_{m,i}^{\text{opt}}\right|^2 = \frac{1}{M}. \tag{45}$$

In this situation, the noise energy is reduced to $1/M$ while the energy of the separated echo[4] is also reduced to $1/M$, which means that the weighting processing will not change SNR.

[4]It should be noted that the pulselength of the preserved signal is reduced to $1/M$.

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In addition, it is no hard to obtain the following equation according to (21):

$$\boldsymbol{W}_{\boldsymbol{m}}^{\mathrm{opt}} = \left[\boldsymbol{w}_{\boldsymbol{1}}^{\mathrm{opt}}, \boldsymbol{w}_{\boldsymbol{2}}^{\mathrm{opt}}, \ldots, \boldsymbol{w}_{\boldsymbol{M}}^{\mathrm{opt}}\right] = \boldsymbol{\Phi}^{-\boldsymbol{1}}. \tag{46}$$

Furthermore, according to (44), (46) can be reformulated, as

$$\frac{1}{M}\boldsymbol{\Phi}^{\boldsymbol{H}} = \boldsymbol{\Phi}^{-\boldsymbol{1}}. \tag{47}$$

Thus, to avoid SNR loss, the matrix $(1/\sqrt{M})\boldsymbol{\Phi}$ should be a unitary matrix (e.g., Fourier matrix).

C. Number of Bandpass Filters

As mentioned before, due to the continuity of signal echoes, bandpass filters should be employed before DBF to ensure that the interference signal segments and the wanted signal segments are associated with certain time intervals. The pulsewidth of the signal segment after filtering decreases with increasing bandpass filter number. In particular, if the number of bandpass filter is sufficiently large, the signal segment will shrink to a single point, which is equivalent to the matched filter. Moreover, the suppression performance of DBF will decline with the increase of suppression areas. Therefore, the suppression performance will improve with the increase of the bandpass filter number. On the other hand, the system cost and complexity will rise with increasing the bandpass filter number. Hence, the number of bandpass filters should be carefully chosen simultaneously considering system cost and interference performance, and further analysis will be presented in the next section.

IV. Simulation Experiment

To verify the practicality of the proposed spaceborne MIMO-SAR solution, the detailed simulation experiments are carried out in this section. Considering a typical spaceborne MIMO-SAR system with two antenna apertures in the azimuth direction, the main parameters are listed in Table I. Two azimuth channels simultaneously transmit two SPC waveforms, respectively, and all antennas simultaneously receive the ground echoes. In addition, Fig. 9 shows an example of the spectra of transmitted SPC waveforms, and the corresponding code phase matrix is

$$\boldsymbol{\Phi} = \begin{bmatrix} e^{j\frac{3}{2}\pi} & e^{j\frac{1}{3}\pi} \\ e^{j\frac{2}{3}\pi} & e^{j\frac{1}{2}\pi} \end{bmatrix}. \tag{48}$$

Notably, the transmitted SPC waveforms are not orthogonal as illustrated in the enlarged portion of the figure.

Fig. 10 shows the received signal at the output of the first and the sixth filter. The yellow and black color signal segments correspond to the interference signal segments of the first and second SPC waveforms, respectively. The red color signal segment is the wanted echo preserved after time-shift weighted processing. In addition, the horizontal blue dotted lines indicate the illumination swath. In both figures, after bandpass filtering, the wanted echo and interference echoes are always separated by $T_s = 20\ \mu$s. Furthermore, in this simulation experiment, after filtering, DBF presented in Section II-C is employed to suppress interferences from other antennas.

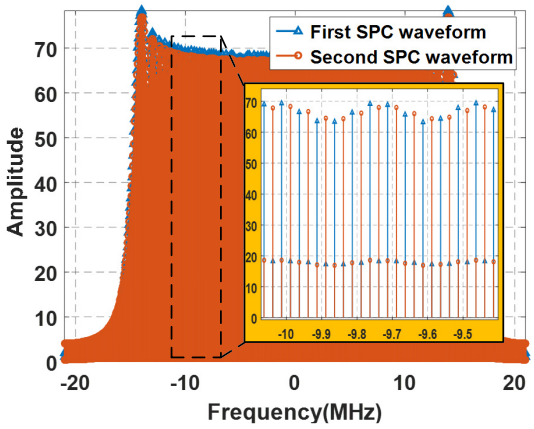


Fig. 9. Spectra illustration of the SPC waveforms.

TABLE I
SPACEBORNE MIMO-SAR SYSTEM PARAMETERS

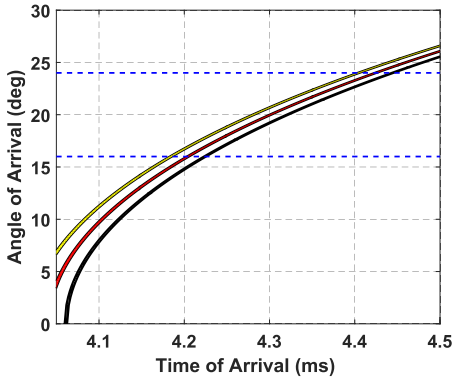
Parameters	Value
Orbit height	607 Km
Look angle	20 deg
Center frequency	9.6 GHz
Pulse duration	40 μs
Bandwidth	30 MHz
PRF	1723 Hz
Antenna length in azimuth	9.6 m
Antenna length in elevation	3 m
Number of channels in azimuth	2
Number of channels in elevation	16
Number of band-pass filters	6
Illumination swath	16-24 deg

An example of such a receive pattern is plotted in Fig. 11. The pattern after bandpass filter No. 4 (in red), at the time of arrival (ToA) of 4.3 ms, has the maximum gain at the corresponding angle of arrival (AoA) of 19.57°. In addition, according to (28), the ToAs of the interference signal segments are 4.3 ms$\pm$20 μs and the corresponding AoAs are 18.8° and 20.3°, and these signal segments are both in the null portion of the DBF pattern (see the yellow and black portion of Fig. 11), thus, the interference energy can be suppressed. Moreover, it should be noted that the waveform type of separated signal segment (in red) depends on the waveform type preserved in the time-shift weighting processing.

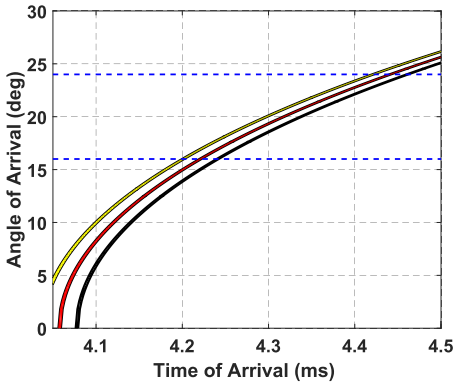
The individual energy contributions after DBF for bandpass filter No. 3 at each time instant are shown in Fig. 12. Notably, compared with the preserved echo energy, the interference echo energy can be suppressed lower -35 dB or more, which is on the order of the cross-pol coupling of the antenna, hence, this suppression rate is sufficient for MIMO-SAR applications.

Furthermore, to verify the effect of bandpass filter number, the interference suppression performance for different numbers of bandpass filters is evaluated by simulations, and the results are shown in Fig. 13. The interference suppression performance improves with increasing the bandpass filter number, which supports the previous analysis. In addition, the interference suppression rate of six bandpass filters is approximately 40.46 dB, which is sufficiently low to meet real system requirements.

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(a)



(b)

Fig. 10. AoAs of the received signal after the (a) first and (b) sixth bandpass filters. The blue dotted lines indicate the illumination swath. The red signal segment is the wanted echo. The yellow and black color signal segments are the interference echoes.

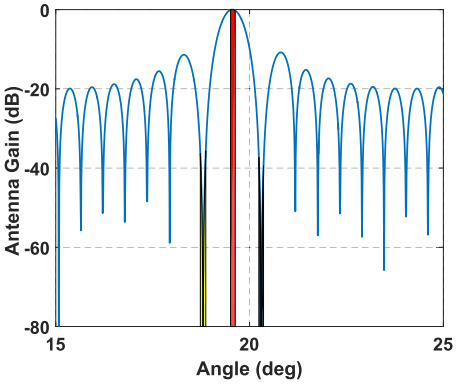


Fig. 11. Antenna pattern after DBF for the bandpass filter No.4 at time 4.3 ms. The wanted signal segment (in red color) is preserved, the interference signal segments (in yellow and black color) are suppressed.

Finally, the distributed scene simulation experiment with the same parameters reported in Table I is performed to further show the advantages and effectiveness of the proposed spaceborne MIMO-SAR imaging scheme. [Fig. 14(a) and (b)] shows a picture of the original scene used to provide the input for the echo simulation of the 1st SPC waveform (2nd SPC waveform). Fig. 14(c) shows the imaging results of the up-down chirp waveforms. It can be clearly seen that the image quality is very poor because the interference energy just spreads throughout the image but does not decrease. However, as shown in Fig. 14(d), the interference energy is greatly

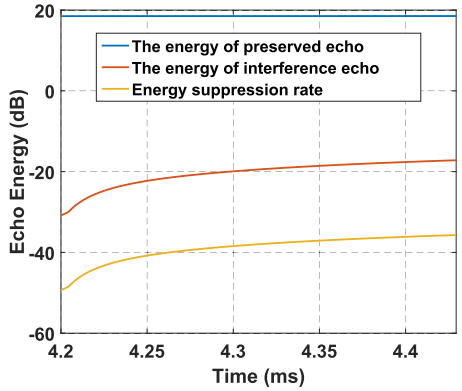


Fig. 12. Energy contributions of different echoes after DBF.

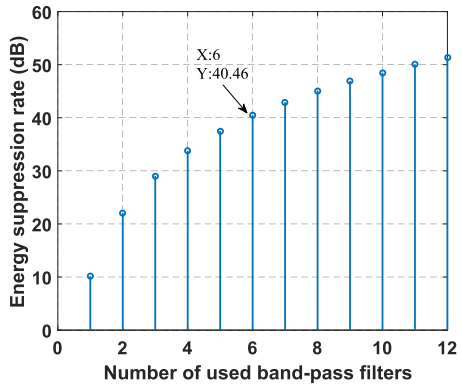


Fig. 13. Interference energy suppression rate for the bandpass filter No. 4 at time 4.3 ms versus different numbers of bandpass filter.

suppressed by the proposed MIMO-SAR solution, and the image quality is close to that for the original scene. In addition, another imaging result for the 2nd SPC waveform is also well focused as shown in Fig. 14(e).

V. Comparisons

A. SPC Waveform Versus OFDM Waveform

In this section, the comparison between the SPC waveform and the OFDM waveform is given. Here, the OFDM waveform are not the conventional OFDM waveform, and it refers to the improved OFDM waveform which avoids the extra carrier frequency deviation. The OFDM waveform discussed in [27] is the conventional OFDM waveform, whereas the OFDM waveform given in [30] are the improved OFDM waveform but it only considers the compensation method for two waveforms. The improved OFDM waveform generation method has been presented in our another article [35]. Here, the mathematical formulations of the conventional OFDM waveform and the improved OFDM waveform are directly given by

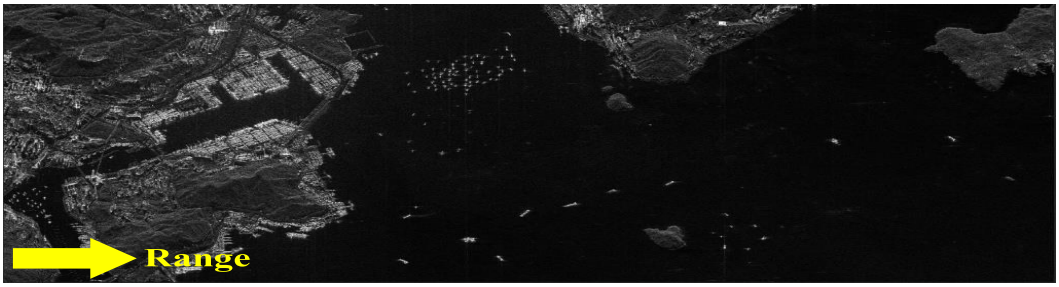
$$s_m^{\text{Conventional}}(t) = e^{j2\pi\frac{m-1}{MT_s}t}\sum_{i=1}^{M} s(t-(i-1)T_s) \tag{49}$$

and

$$s_m^{\text{Improved}}(t) = \sum_{i=1}^{M} e^{j\frac{2\pi}{M}(i-1)(m-1)} s(t-(i-1)T_s) \tag{50}$$

respectively.

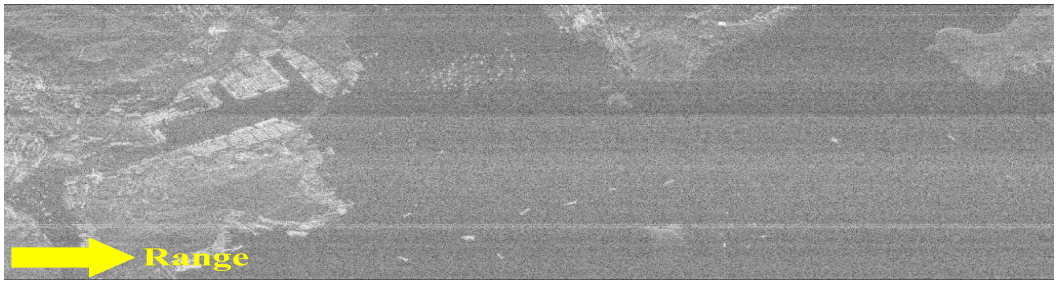
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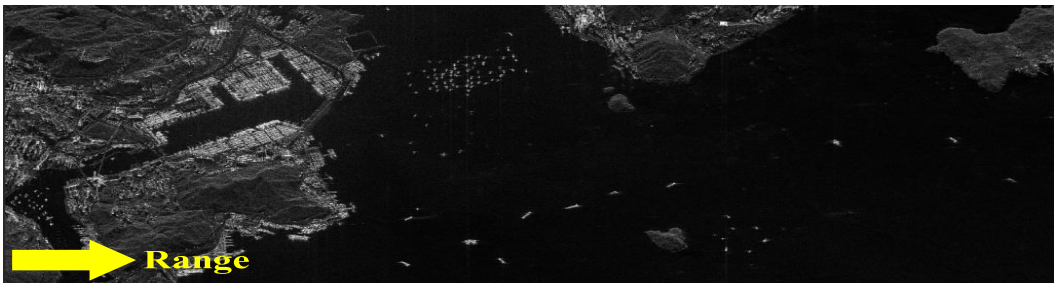
(a)



(b)



(c)



(d)



(e)

Fig. 14. Distributed scene simulation results of MIMO-SAR with 2 transmit antennas. The horizontal direction is the range direction and the vertical direction is the azimuth direction. (a) Original scene illuminated by the 1st SPC waveform. (b) Original scene illuminated by the 2nd SPC waveform. (c) Imaging results of the up-down chirp waveforms. (d) Imaging results of the separated echo (1st SPC waveform) by the proposed method. (e) Imaging results of the separated echo (2nd SPC waveform) by the proposed method.

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Waveform	Phase code matrix $\frac{1}{\sqrt{M}}\Phi$
Phase code modulated waveform	Arbitrary matrix
SPC waveform for the echo separation	Invertible matrix
SPC waveform for a maximum SNR	Unitary matrix
Orthogonal phase code modulated waveform	Unitary matrix
OFDM waveform	$\Phi_{m,n} = e^{j\frac{2\pi}{M}(n-1)(m-1)}$

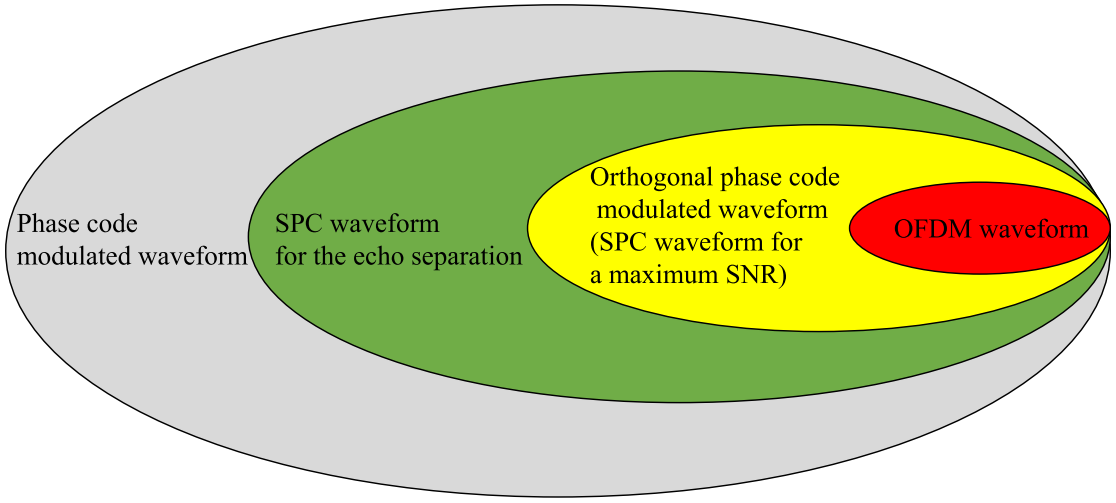


Fig. 15. Connection between the phase code modulated waveform, SPC waveform, OFDM waveform and orthogonal phase modulated waveform.

As shown in (49), the conventional OFDM waveforms have a small extra carrier frequency deviation ($(m-1)/MT_s$), but this term is avoided in (50). Moreover, it can be seen that the mathematical form of the improved OFDM waveform is similar to that of the SPC waveform, and the method proposed in this article is inspired by the improved OFDM waveform. Therefore, it is necessary to discuss the relationship between the SPC waveform and the OFDM waveform. Furthermore, the phase code matrix of the improved OFDM waveform is

$$\boldsymbol{\Phi}_{m,n} = e^{j\frac{2\pi}{M}(n-1)(m-1)}. \tag{51}$$

Letting $\boldsymbol{E} = (1/M)\boldsymbol{\Phi}\boldsymbol{\Phi}^H$, $\boldsymbol{E}_{m,n}$ can be calculated as

$$\begin{aligned}\boldsymbol{E}_{m,n} &= \frac{1}{M}\sum_{k=1}^{M} e^{j\frac{2\pi}{M}(n-1)(m-1)} e^{-j\frac{2\pi}{M}(n-1)(m-1)} \\ &= \frac{1}{M}\sum_{k=1}^{M} e^{j\frac{2\pi}{M}(n-m)(k-1)}.\end{aligned} \tag{52}$$

If $n \neq m$, (52) can be calculated as

$$\boldsymbol{E}_{m,n} = \frac{1}{M}\frac{1-1}{1-e^{j\frac{2\pi}{M}(n-m)}} = 0. \tag{53}$$

If $n = m$, (52) can be calculated as

$$\boldsymbol{E}_{m,n} = \frac{1}{M}\boldsymbol{E}_{n,n} = \frac{1}{M}\sum_{k=1}^{M} 1 = 1. \tag{54}$$

Therefore, $\boldsymbol{E}$ is a unit matrix, which means the OFDM waveform is just a special case of the SPC waveform. Moreover, Fig. 15 provides a graphical interpretation of the relation between the phase code modulated waveforms, SPC waveforms, OFDM waveforms, and the orthogonal phase code modulated waveforms. The phase code matrix of OFDM waveforms should satisfy (51) and it is a kind of orthogonal phase code modulated waveforms. If ignoring the SNR performance, the requirement of phase code matrix for SPC waveforms is reversibility; thus, the orthogonal phase code modulated waveform is a kind of SPC waveform. Furthermore, to obtain the maximum SNR, SPC waveforms will become orthogonal phase code modulated waveforms.

B. Comparisons Between Different MIMO-SAR Schemes

In this section, comparisons between the proposed scheme, the STC schemes and the orthogonal-waveform beamforming schemes are detailed.

The transmission scheme of the STC MIMO-SAR system is shown in Fig. 16(a), and the signal model of STC MIMO-SAR system can be formulated as [26]

$$\begin{aligned}&\boldsymbol{R_n}(f_t, f_\eta) \\ &= \begin{bmatrix} R_{n,1}(f_t, f_\eta) \\ R_{n,2}(f_t, f_\eta) \\ \vdots \\ R_{n,K}(f_t, f_\eta) \end{bmatrix} \\ &= \boldsymbol{S}(f_t, f_\eta)\boldsymbol{H_n}(f_t, f_\eta) + \boldsymbol{N_n}(f_t, f_\eta) \\ &= \begin{bmatrix} S_{1,1} & S_{1,2} & \cdots & S_{1,M} \\ S_{2,1} & S_{2,2} & \cdots & S_{2,M} \\ \vdots & \vdots & \ddots & \vdots \\ S_{K,1} & S_{K,2} & \cdots & S_{K,M} \end{bmatrix} \begin{bmatrix} H_{1,n} \\ H_{2,n} \\ \vdots \\ H_{M,n} \end{bmatrix} + \begin{bmatrix} N_{1,n} \\ N_{2,n} \\ \vdots \\ N_{M,n} \end{bmatrix}\end{aligned} \tag{55}$$

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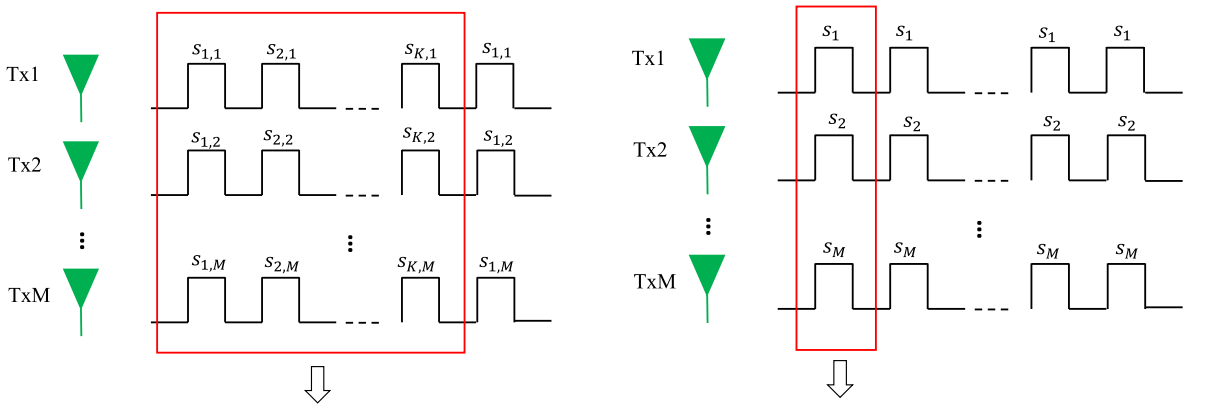


Fig. 16. Illustration of transmission scheme of the STC scheme and the proposed method. (a) STC scheme. (b) Proposed method.

where $R_{n,i}$ is the echo signal of the nth receive antenna at the ith Pulse Repetition Interval (PRI), $S_{i,j}(f_t, f_\eta)$ denotes the transmitted waveform of the jth transmit antenna at the ith PRI, and $H_{i,n}(f_t, f_\eta)$ $(N_{i,n}(f_t, f_\eta))$ is the channel response (noise) corresponding to the ith transmit antenna and the nth receive antenna. A decoding matrix $\boldsymbol{D}$ is employed to separate the echoes, and it should satisfy

$$\boldsymbol{D} \cdot \boldsymbol{S} = \boldsymbol{I_M}. \tag{56}$$

The necessary conditions of (56) is $\mathrm{Rank}(\boldsymbol{D}) = \mathrm{Rank}(\boldsymbol{S}) = M$ and $K \geq M$ [36], which means the PRF should be larger than MB_a (B_a is the Doppler bandwidth). For a visualized comparison, the transmission scheme of the proposed method is shown in Fig. 16(b), and the method of this article does not need to use multiple PRI data to obtain one PRI separated echo.

The basic idea underlying the orthogonal-waveform beamforming schemes can be expressed as a restricted short-term shift-orthogonality condition [7], namely,

$$\int h_{\mathrm{DBF}}(\tau) \cdot s_i(t) \cdot s_j^*(t-\tau)dt = 0 \quad \forall \tau \in \mathbb{R},\ i \neq j \tag{57}$$

where $s_i(t)$ is the waveform of the ith transmitted antenna, and $h_{\mathrm{DBF}}(\tau) = \mathrm{rect}(\tau/\Delta\tau)$ is a spatial filter. The role of DBF is to suppress the interferences from the far arrival angles, while the transmitted waveforms should be orthogonal in $\Delta\tau$ for separating the echoes from the close arrival angles. However, in the proposed method, the requirement of orthogonality in $\Delta\tau$ is replaced by a simple time-shift weighting processing.

C. Further Discussion Based on Functional Analysis Theory

In this section, for better showing the main idea and contributions of this article, based on the discussion in the Section V-B, some new insights can be gained by the functional analysis theory.

MIMO-SAR system can be defined as an Abstract Algebra System (AAS), and the MIMO-SAR echo separation processing can be defined as an Operator for this AAS

$$\Gamma_{\boldsymbol{\Omega}}: \begin{cases} \Gamma_{\boldsymbol{\Omega}}\{\bar{s}_i(\boldsymbol{\omega}; \eta; \tau; f_D), h_j(\boldsymbol{\Omega})\} = \boldsymbol{0} \\ \qquad \forall i \neq j, \quad \tau \in \mathbb{T}_\tau, \quad f_D \in \mathbb{F}_D \\ \Gamma_{\boldsymbol{\Omega}}\{\bar{s}_i(\boldsymbol{\omega}; \eta; \tau; f_D), h_j(\boldsymbol{\Omega})\} = \chi_i(t; \eta; \tau; f_D) \\ \qquad \forall i = j, \quad \tau \in \mathbb{T}_\tau, \quad f_D \in \mathbb{F}_D \end{cases} \tag{58}$$

where $i, j = 1, \ldots, M$ (M refers M echoes), τ (f_D) denotes the echo delay (Doppler frequency) referenced to the time of closest approach, respectively, $\mathbb{T}_\tau$ and $\mathbb{F}_\eta$ are the corresponding spaces, which can be represented as

$$\begin{aligned} \mathbb{T}_\tau &= [\tau^{\min}, \tau^{\max}], \quad \mathbb{T}_\tau \subseteq \mathbb{R} \\ \mathbb{F}_D &= \left[f_\eta^{\min}, f_\eta^{\max}\right], \quad \mathbb{F}_D \subseteq \mathbb{R} \end{aligned} \tag{59}$$

where $\tau^{\min}$ and $f_\eta^{\min}$ denote the minimum echo delay and Doppler frequency, respectively, $\tau^{\max}$ and $f_\eta^{\max}$ denote the maximum echo delay and Doppler frequency, $\bar{s}_i(\boldsymbol{\omega}; \eta; \tau; f_D)$ is the echo corresponding to the ith transmitted signal $s_i(\boldsymbol{\omega})$, $h_j(\boldsymbol{\Omega})$ is the generalized matched function for the jth echo, $\chi_i(t; \eta; \tau; f_D)$ is the generalized matched output function of the ith echo, $\boldsymbol{\omega}$ denotes the domain of definition for the transmitted waveform, $\boldsymbol{\Omega}$ is the processing domain for the Operator $\Gamma_{\boldsymbol{\Omega}}$, and $\boldsymbol{\omega} \subseteq \boldsymbol{\Omega}$. SAR transmits radar signals to extract the delay and Doppler frequency information included in the echo, in other word, the Operator $\Gamma_{\boldsymbol{\Omega}}$ is performed in $\boldsymbol{\Omega}$ to obtain the ground information τ and f_D. The spaces of transmitted waveforms $\mathbb{S}$ and matched functions $\mathbb{H}$ are

$$s_i \in \mathbb{S}, \quad h_i \in \mathbb{H}; \ \mathbb{S}, \mathbb{H} \subseteq \boldsymbol{L}^2 \tag{60}$$

respectively.

First, considering the traditional single-input single-output SAR, the transmitted waveforms are the functions of the fast time t, and $\Gamma_{\boldsymbol{\Omega}}$ is the 2-D time correlation. In this respect, $\boldsymbol{\omega} = t$, $\boldsymbol{\Omega} = (t, \eta)$. Moreover, SAR is an imaging radar, hence, $\mathbb{S}$ is the space of the waveforms suiting for SAR imaging (e.g. LFM waveform and NLFM waveform), which is denoted by $\mathbb{S}_{\mathrm{SAR}}$.

Next, considering an MIMO-SAR system, for the orthogonal-waveform schemes, the transmitted waveforms are

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TABLE II
COMPARISON BETWEEN DIFFERENT MIMO-SAR SCHEMES

Method	Main idea of echo separation	Advantages	Drawbacks
STC scheme proposed in [21]	Separated in the slow time domain by a coding and decoding matrix	1) No requirement for DBF	1) PRF $> MB_a$ 2) Residual ambiguous caused by the time-variant channel response
STC scheme proposed in [24]	Separated in the range-Doppler domain by a coding and decoding matrix	1) No requirement for DBF 2) Avoid the effect of the time-variant channel response	1) PRF $> MB_a$ 2) SNR loss
STC scheme proposed in [26]	Separated in the range-Doppler domain by a coding and decoding matrix	1) No requirement for DBF 2) Avoid the effect of the time-variant channel response 3) Avoid the SNR loss in [24]	1) PRF $> MB_a$
OFDM waveform beamforming schemes proposed in [27], [30]	OFDM waveforms, and DBF on receive	1) No requirement for the increase of PRF 2) Avoid the effect of the time-variant channel response	1) Need DBF 2) Suffer from the effect of Doppler leakage
STSO waveform beamforming schemes proposed in [7], [29]	STSO waveforms, and DBF on receive	1) No requirement for the increase of PRF 2) Avoid the effect of the time-variant channel response 3) Avoid the effect of Doppler leakage	1) Need DBF 2) Need to meet short-term shift-orthogonality condition
The method proposed in this paper	Time-shift weighting process and DBF on receive	1) No requirement for the increase of PRF 2) No requirement for the orthogonality of transmitted waveform 3) Avoid the effect of the time-variant channel response 4) Avoid the effect of Doppler leakage	1) Need DBF

the function of the fast time t, and $\Gamma_{\mathbf{\Omega}}$ is also the 2-D time correlation. However, the transmitted waveform space is

$$\left\{ s_i | s_i \in \mathbb{S}_{\mathrm{SAR}}, \int s_i(t) s_j^*(t-\tau) = 0, \forall \tau \in \mathbb{R}, i \neq j \right\}. \quad (61)$$

However, for the waveforms with an overlapped spectral support, the orthogonality for arbitrary time shifts cannot be hold.

For the STC schemes, the transmitted waveforms are the functions of the fast time t and the slow time η. The MIMO-SAR separation processing can be performed in the Doppler frequency domain, see [26]. In this respect, $\boldsymbol{\omega} = (t, \eta)$, $\mathbf{\Omega} = (t, \eta, f_D)$, $\Gamma_{\mathbf{\Omega}}$ is the diagonalization for the echo matrix (namely, (56)), and the transmitted waveform space is $\mathbb{S}_{\mathrm{SAR}} \times \mathbb{M}_\eta$, where $\mathbb{M}_\eta$ is the azimuth coding space (e.g., Alamouti code), and "$\times$" is the Cartesian Product. On the other hand, a part of Doppler information is used for echo separation, hence, these Doppler information is lost. For example, half Doppler information is lost for an MIMO-SAR system with two transmit antennas, leading a decrease of PRF.

For the STSO orthogonal beamforming schemes, a new observation variable (namely, the direction of arrival φ) is introduced by the DBF. Correspondingly, $\boldsymbol{\omega} = t, \mathbf{\Omega} = (t, \eta, \varphi)$ and $\Gamma_{\mathbf{\Omega}}$ is the weighting correlation. In the STC schemes, Doppler information is used for MIMO-SAR separation, whereas it is also the echo information we want to obtain; hence, it will reduce the observation performance of radar. In the orthogonal-waveform beamforming schemes, φ information is used, hence, this information is lost in the final output. Fortunately, φ is not the information we want. On the other hand, the observation accuracy of φ is limited by the channel number in elevation. Therefore, STSO waveforms are proposed to ensure the orthogonality of echoes whose φ cannot be distinguished by DBF, and the transmitted waveform space

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is

$$\left\{ s_i | s_i \in \mathbb{S}_{\mathrm{SAR}}, \int h_{\mathrm{DBF}}(\tau) \cdot s_i(t) \cdot s_j^*(t-\tau) dt = 0, \quad \forall \tau \in \mathbb{R}, i \neq j \right\}. \quad (62)$$

For the proposed method, a new variable (namely, phase code ϕ) is introduced by the phase code matrix, and it should be noted that ϕ is not a new observation variable. The role of ϕ is to relieve the short-term shift-orthogonality restriction. Correspondingly, $\boldsymbol{\omega} = (t, \phi)$, $\boldsymbol{\Omega} = (t, \phi, \eta, \varphi)$, and the transmitted waveform space is $\mathbb{S}_{\mathrm{SAR}} \times \mathbb{C}_\phi$ with the optimal phase code space[5]

$$\mathbb{C}_\phi = \left\{ \phi | \phi \in \mathbb{C}, \text{ The phase code matrix } \frac{1}{\sqrt{M}} \boldsymbol{\Phi} \text{ is a unitary matrix} \right\}. \quad (63)$$

In this respect, the optional waveform space is extended to $\mathbb{S}_{\mathrm{SAR}}$ compared with the orthogonal-waveform beamforming schemes, and the corresponding Operator $\Gamma_{\boldsymbol{\Omega}}$ is the time-shift weighting beamforming processing.

The design of radar waveforms is a key factor affecting radar performance which has been extensively studied, and a lot of advance radar waveforms are proposed for improving the radar performance. Generally, these radar waveforms cannot meet the short-term orthogonality condition. Admittedly, for some simple waveforms (e.g., LFM waveform), it is easy to construct the corresponding STSO waveform by a simple processing, while keeping waveform performance. However, for the complex waveforms with a high imaging performance (see [32], [33], [37]), the short-term orthogonality condition is an additional design goal, which will increase the complexity of the waveform design and further lose the waveform performance. The method proposed in this article can relieve this restriction, which can let all advanced radar waveforms suiting for SAR can be directly employed in the MIMO-SAR system to improve the radar performance, e.g., SNR improvement, sidelobes suppression.

On the other hand, as discussed in the Section II-C, DBF on receive should enable a reliable separation of the echo signals with a T_s time delay difference, and the requirement for DBF is the same with the STSO waveform beamforming scheme, which means the antenna height of an optimally oriented antenna h_{ant} should exceed [7]

$$h_{\mathrm{ant}} \geq \frac{2\lambda R_{\max} \tan(\theta_{\mathrm{inc,\,max}})}{cT_s} \quad (64)$$

where $R_{\max}$ is the far range distance and $\theta_{\mathrm{inc,\,max}}$ is the maximum incident angle. The comparisons between different MIMO-SAR schemes are given in Table II.

VI. Conclusion

In this article, the SPC waveform, which is a kind of phase code modulated waveform, is introduced as the transmitted waveform for solving the echo separation problem associated with MIMO-SAR systems. Compared with the state-of-the-art STSO waveform beamforming schemes, this scheme relieves the short-term shift-orthogonality condition, which leads to an expanded optional waveform space. The role of STSO waveforms, which is to ensure the separation of the echoes from the close arrival angles, is replaced by a simple time-shift weighting processing. Furthermore, a range bandpass filter bank and DBF technique are employed to ensure that the echoes from the far arrival angles are separable. Compared with the other existing MIMO-SAR schemes, the proposed method can avoids the effects of Doppler leakage and the time-varying channel response, and it does not need to increase PRF. At the analysis stage, the detailed numerical simulation experiments are employed to verify the feasibility of the proposed scheme. Finally, the relationship of different waveforms and the performance comparisons between different MIMO-SAR schemes are detailed.

[5]To obtain a maximum SNR, $(1/\sqrt{M})\boldsymbol{\Phi}$ should be a unitary matrix, otherwise, it just needs to be reversible.

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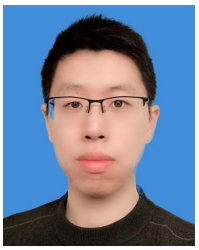
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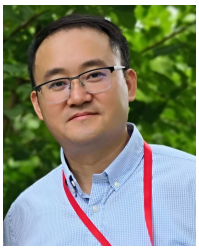
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